

Distance between two lines

Basic skills

Q1. Skew lines are non-parallel lines that do not intersect

Q2. Difference is $\begin{pmatrix} 6 \\ 2 \\ -1 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ -3 \end{pmatrix}$

so distance $\sqrt{4^2 + 1^2 + 3^2} = \sqrt{26}$

Q3.

a) $\underline{r} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$

b) need another point so $\lambda = 1 \quad \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$

Then $\left(\underline{r} - \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}\right) \times \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} = \underline{0}$

Q4. $\underline{r}_1 = \underline{r}_2 \Rightarrow \begin{pmatrix} 2+3t \\ 1+t \\ -3-t \end{pmatrix} = \begin{pmatrix} s \\ 1+s \\ -2 \end{pmatrix}$

$$2+3t=s$$

$$1+t=1+s \Rightarrow t=s \text{ so } 2+2t=0 \Rightarrow t=-1=s$$

then $-3 - -1 = -2$ so consistent so $\begin{pmatrix} -1 \\ 0 \\ -2 \end{pmatrix}$

Competency skills

Q5. $\underline{r}_1 = \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$, $\underline{r}_2 = \begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$

a) $\underline{n} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 0 & 1 \\ 3 & -1 & 2 \end{vmatrix} = \underline{i} - \underline{j} - 2\underline{k}$
 $= \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$

b) $\frac{\left| \left(\begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix} \right) \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \right|}{\sqrt{1^2 + 1^2 + 2^2}} = \frac{\left| \begin{pmatrix} 2 \\ -3 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \right|}{\sqrt{6}}$
 $= \frac{|2 + 3 + 10|}{\sqrt{6}} = \frac{15}{\sqrt{6}}$

Q6. $\underline{n} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix} = -\underline{i} + 7\underline{j} + 5\underline{k}$

so $\frac{\left| \left(\begin{pmatrix} -5 \\ 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right) \cdot \begin{pmatrix} -1 \\ 7 \\ 5 \end{pmatrix} \right|}{\sqrt{1^2 + 7^2 + 5^2}} = \frac{|5 + 14 + 5|}{\sqrt{75}} = \frac{24}{5\sqrt{3}}$

Q7. As $-6 \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} = \begin{pmatrix} -12 \\ 6 \\ -30 \end{pmatrix}$ parallel lines so

pick a point $\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$ and distance to Γ_2

Point to
line

$$\text{so } \left(\Gamma_2 - \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} \right) \cdot \begin{pmatrix} -12 \\ 6 \\ -30 \end{pmatrix} = 0$$

scale down for ease

$$\Rightarrow \begin{pmatrix} -12\mu - 2 \\ 6\mu - 2 \\ -30\mu + 11 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} = 0$$

$$\Rightarrow -24\mu - 4 - 6\mu + 2 - 150\mu + 55 = 0$$

$$\Rightarrow -180\mu = -53$$

$$\Rightarrow \mu = \frac{53}{180}$$

so vector to line is $\begin{pmatrix} -12 \times \frac{53}{180} - 2 \\ 6 \times \frac{53}{180} - 2 \\ -30 \times \frac{53}{180} + 11 \end{pmatrix}$

$$= \frac{1}{180} \begin{pmatrix} -996 \\ -42 \\ 390 \end{pmatrix} \text{ so distance is}$$

$$\frac{1}{180} \sqrt{996^2 + 42^2 + 390^2} \approx 5.95$$



Q 8. $A = (0, 1, 0)$, $B = (5, 3, 2)$
 $C = (-1, 2, 3)$, $D = (4, 2, -4)$

a) AB direction $\vec{AB} = \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix}$

so $\vec{r} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix}$

CD direction $\vec{CD} = \begin{pmatrix} 4 \\ 2 \\ -4 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ -7 \end{pmatrix}$

so $\vec{r} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 0 \\ -7 \end{pmatrix}$

b) $\vec{AB} \times \vec{CD} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 5 & 2 & 2 \\ 5 & 0 & -7 \end{vmatrix} = -14\underline{i} + 45\underline{j} - 10\underline{k} = \begin{pmatrix} -14 \\ 45 \\ -10 \end{pmatrix}$

c) $\frac{\left| \left(\begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right) \cdot \begin{pmatrix} -14 \\ 45 \\ -10 \end{pmatrix} \right|}{\sqrt{14^2 + 45^2 + 10^2}} = \frac{|14 + 45 - 30|}{\sqrt{2321}} \approx 0.602$

Q9. $3x - y + z = 2$, $2z - x = 5$

So for plane intersection

$$x = 2z - 5$$

$$6z - 15 - y + z = 2 \Rightarrow 7z - y = 17$$

$$y = 7z - 17 \quad z = \frac{y + 17}{7}$$

$$x = 2z - 5 = \frac{2y - 1}{7}$$

so $y = \frac{7x + 1}{2}$

$\Rightarrow \frac{7x + 1}{2} = y = 7z - 17$ is line

so $\frac{x + \frac{1}{7}}{2} = \frac{y}{7} = z - \frac{17}{7}$

and $y = 3x$, $z = 2 \Rightarrow \frac{y}{3} = x$

directions are $\begin{pmatrix} 2 \\ 7 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$

For points $(-\frac{1}{7}, 0, \frac{17}{7})$, $(0, 0, 2)$

$$\underline{n} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 7 & 1 \\ 1 & 3 & 0 \end{vmatrix} = -3\underline{i} - \underline{j} - \underline{k} = \begin{pmatrix} -3 \\ -1 \\ -1 \end{pmatrix} \text{ use } \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$$

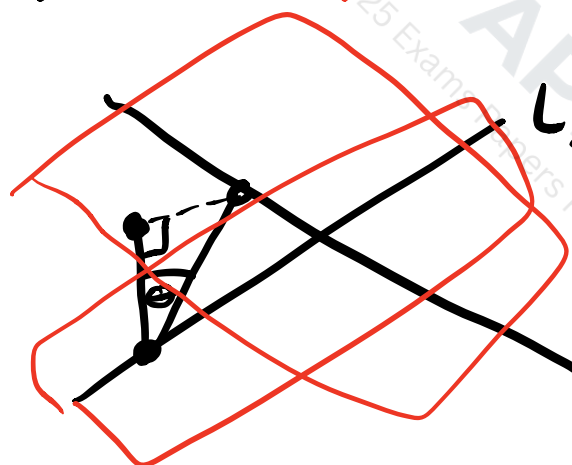
Then formula

$$\frac{\left| \left(\begin{pmatrix} -\frac{1}{7} \\ \frac{10}{7} \\ \frac{17}{7} \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} \right) \cdot \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \right|}{\sqrt{3^2 + 1^2 + 1^2}} = \frac{\left(-\frac{3}{7} + \frac{3}{7} \right)}{\sqrt{11}} = 0$$

So lines intersect and aren't skew/
parallel

Q10.

Planes containing lines



$$L_2: \underline{r} = \underline{a}_2 + \mu \underline{d}_2$$

$$L_1: \underline{r} = \underline{a}_1 + \lambda \underline{d}_1$$

Planes

so

$$\begin{aligned} \text{dist} &= |\underline{a}_2 - \underline{a}_1| \cos \theta \\ &= |\underline{a}_2 - \underline{a}_1| \frac{(\underline{a}_2 - \underline{a}_1) \cdot \underline{n}}{|\underline{a}_2 - \underline{a}_1| |\underline{n}|} \\ &= \frac{(\underline{a}_2 - \underline{a}_1) \cdot \underline{n}}{|\underline{n}|} \end{aligned}$$

$$\cos \theta = \frac{(\underline{a}_2 - \underline{a}_1) \cdot \underline{n}}{|\underline{a}_2 - \underline{a}_1| |\underline{n}|}$$

need mod as
want positive
distance

$$\text{so } \frac{|(\underline{a}_2 - \underline{a}_1) \cdot \underline{n}|}{|\underline{n}|}$$